PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

POSTGRADUATE DIPLOMA IN LIBRARY AUTOMATION AND NETWORKING EXAMINATIONS, 2012

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR POSTGRADUATE DIPLOMA IN LIBRARY AUTOMATION AND NETWORKING (PGDLAN) FOR THE EXAMINATIONS, 2012.

Paper Code	Title of the Paper	Marks			
		Internal Assessment*	Ext. Exam.	Total	Exam Hours
LAN- 01	Basics of Computers and Communication	20	80	100	3
LAN- 02	Library Automation	20	80	100	3
LAN- 03	Information Systems	20	80	100	3
LAN- 04	Networking and Internet Technology	20	80	100	3
LAN- 05	Internet Resources and Content Development	20	80	100	3
LAN- 06	Computer Practical	20	80	100	3

* For the purpose of Internal Assessment in all papers two assignments will be given (10 marks each), which will be compulsory**.**

Paper Code: LAN-01:
Paper Title: Basics of Computers and Communication

Objectives: To provide overall basic knowledge of Computers and also the concept of Hardware and Software.

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Unit-1

Hardware

Need for Computerization of Libraries: Library Automation, Need/Purpose/Objectives of Library Automation, Areas where Library Automation is desirable. **Computers and their Components:** Computer Fundamentals- definition, characteristics, functional units, input devices, output devices, secondary storage devices, central processing unit. Software - types of software. **Storage Media-** RAM, ROM, Flash Memory, Blue-Ray Disk.

Unit-2

Software

Introduction to Operating System: Translators, Operating System- Functions, types.
Operating Systems: Dos, Windows, Unix and Linux
Application Software: Using Windows, MS-Access, MS-Excel, and MS-Power Point

Unit-3

Computers and Society

National Information Infrastructure (NII): Basics of NII, Technological features, Components, GII, Impact of NII on Library and Information Service
Information Society: Definitions, Development of Information Society Model, Impact of Information Society on Information Profession, Concept of Information Society in India

Unit-4

Communications

Elements of Data Communication: Basics of Data Communication, Telecommunication Network, Transmission Signals, Communication Channels, Medium, and Hardware.

Essential Readings

Bingham,J.(1989) Data Processing. Basingstoke:Macmillan.

Bissmer, R.H. (1993). Introduction to Computer Concepts. New York: John Wiley.

Jain, V.K. (2000) 'O' Level- Information Technology: BPB.

Jaiswal, S(1999).Information Technology today. New Delhi Galgotia.

Kimber, R.T. (1974). Automation in Libraries. 2nd edition. Oxford: Perabamon.

Rajaraman, V (1997). Fundamentals of Computers. 2nd ed. New Delhi: Prentice Hall of India.

Rowley (1997). Electronoc Library.4th ed.London: Library Association.

Tedd, Lucy A. (1993). An Introduction to Computer based Library Systems. 3rd ed. Chichester:Wiley.

Further Readings

Ericson, Gary (2001) Microsoft Office 2000 Resources Kit.- New Delhi Prentice Hall of India.

Kerninghan, Brian and Pike, Rob (1984). The UNIX programming Environment. _ New Jersey: Prentice-Hall.

Stroel, Stefan and Ulil, Thomas (1996). Linux Universe: Installation and Configuration.-2nd ed.- Heil bronn:Springer

Tanenbaum, Andrew S. and Woofhull, Albert S.(2000). Operating Systems Design and Implementation: New Delhi: Prentice Hall of India.

Paper Code: LAN-02
Paper Title: Library Automation

Objectives: To provide information about latest concepts of Library Automation especially different Library Software.

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Unit -1

Introduction to Library Automation

Historical Perspective: Hays' Model for the History of Library Automation, History of Library Automation in India, Stages of Library Automation, Need and Purpose, Library Automation Trends.

Approaches to Library Automation, Advantages and Barriers of Library Automation

Library Security: CCTV, RFID

Unit-2

Management of Library Automation

Planning of Library Automation: various steps

Retrospective Conversion: Need for Retrospective Conversion, Standards for Retro conversion, various options in retrospective conversion

Formats and Standards: Historical perspective, MARC, UNIMARC, CCF, Bibliographic standards and the internet

Implementation and Evaluation: Library automation implementation and evaluation

Unit-3

Library Software

SOUL: Introduction, features, standards, modules and pricing policy

KOHA: Introduction, features, modules and installation of KOHA on LINUX & WINDOWS platform.

NewGenLib: Introduction, features, modules and installation of NewGenLib on LINUX & WIDOWS platform.

Unit-4

Computerized Information Services

Reference Services: Definitions of reference services, components and functions, levels, need for reference services, types of reference services, reference question, Digital reference services: advantages, models, forms

Document Delivery Services: Document delivery system, electronic document delivery service, software for EDD, methods of EDD, advantages and disadvantages of EDD

Bibliographic Services: Types of bibliography, bibliographic database, and database search interfaces, bibliographic databases on the www

Alerting Services: Current Awareness Services, Selective Dissemination of Information

Essential Readings

Duval, B.K. And Main, L. (1992) Automated Library Systems: A Librarian's Guide and Teaching Manual. London: Meckler.

Kimber, R.T. (1970) Automation in Libraries. Oxford: Pergamon Press.

Kumar, P.S.G. (1987). Computerization of India Libraries. Delhi: B.R. Publishing House.

Pitkin, G.M. (ed.) (1991). The Evolution of Library Automation: Management Issues and Future Perspectives. London: Meckler.

Riaz, Muhammad (1995). Library Automation. New Delhi: Atlantaic.

Roce, James (1994) Introduction to Library Automation. Littleton: Libraries Unlimited.

Rowley, J.E. (1980) Computers for Libraries . 2nd ed. London: The Library Association.

Sharma Pandey, S.K. (1995).Fundamentals of Library Automation. New Delhi: Ess Ess Publications.

Further Readings

Bierman, K.J. (1974). Library Automation. In: Annual Review of Information Science and Technology. Vol.9 Washington: American Society for Information Science.

Saffady, William (1989) Library Automation_ An Overview. Library Trends. 37(3), 269-81.

Salemon, Stephen. Library Automation. In: Encyclopedia of Library and Information Science. Vol 14 New York: Marcel Dekker.

Vashishth, C.P. (ed.) (1988). Modernasitation: in Indian Libraries. Delhi: Indian Library Association.

Paper Code: LAN-03
Paper Title: Information Systems

Objectives: To provide basic knowledge of information storage, processing and retrieval.

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Unit-1

System Analysis and Design

General Systems Theory: History and developments in general systems theory, system's approach, Information systems: components and types, Information systems in Libraries

Information System Design and Development: System analysis, system design, system development methodology, systems development life cycle (SDLC), system implementation, evaluation and documentation, system description techniques

Unit-2

Type of Information Systems

Information Retrieval Systems: Information retrieval architectures, categories of information retrieval systems, purpose, components, functions and models of information retrieval systems, evaluation of Information retrieval system

Knowledge Based Retrieval Systems: Expert system; need, components or structure, operation of expert system, construction of expert system

Management Information Systems and Decision Support Systems: Concept and definition of MIS, need, structures, design and development of MIS, implementation, evaluation and maintenance of MIS, MIS and library and information centers; DSS: concept and definition, structure of DSS

Database Management Systems: Definition and concept, DBMS languages, history and need of DBMS, DBMS components, development, trends, advantages and disadvantages, data models

Unit-3

Data Models and Database Design

Data Structures and File Organization: Concept, types of data structures; file organization techniques, models for logical data organization

Data Models: Data modeling process, types of data models

Database Systems and Architecture: Three-schema architecture

Unit-4

Query Language and Query Processing

Query Language: Structured query language, features of SQL, SQL commands

Boolean Logic: Search strategy, developing effective search strategy, Boolean search

Structured Query Formulation: Query optimization, cost based optimizers

Essential Readings

Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, (2008) *Introduction to Information Retrieval*, Cambridge University Press.

Doyle, L.B. (1975) Information Retrieval and Processing, Melville, Los Angeles

Foskett, A.C.. (1996) The subject approach to information 5th ed., by London: Library Association Publishing.

Hicks, James O. (1993). Management Information Systems: A User Perspective. Minneapolis: west.

Holsapple, C.W., and A. B. Whinston. (1996). Decision Support Systems: A Knowledge-Based Approach. St. Paul: West Publishing.

Lucas, **Henry C.** (1985). The analysis, design, and implementation of information systems. New York: McGraw-Hill.

McLeod, Raymond and Schell George (2000). Management Information Systems (8th Edition) NJ: Prentice Hall.

Waterman, D.A (1996) A Guide To Expert Systems Addison-Wesley Publishing Company.

Further Readings

Davis Gordon B, ed. (1999). The Blackwell Encyclopedic Dictionary of Management Information Systems. Oxford: Blackwell.

Deanna B. Marcum Digital Libraries: For Whom? For What? The Journal of Academic Librarianship, Volume 23, Number 2, March 1997.

Kenny AR and Chapman S (1996). Digital imaging for libraries and archives NY: Cornell

Noerr, P (2000). The Diigtal Library tool kit. Palo Alto: Sun System.

Patter A (1994) LIS professional as knowledge engineers NJ: Learned.

Paper Code: LAN-04
Paper Title : Networking and Internet Technology

Objectives: To provide basic knowledge of networks at macro level and to study detailed information about Internet.

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Unit- 1

Introduction to Networks

Types of Networks: Uses and advantages of networking, types of networks, network topologies, network components

Network Technology: OSI model, Virtual Private Network (VPN), Internet; how internet works, history of internet, Organizations associated with Internet control

Data Networks: Data transmission and signaling, ISDN, DSL, ATM

Network Softwares: Network architecture and role of network softwares, network operating systems, network management system and protocols

Unit-2

Library and Information Networks

Resource Sharing Technology: Need of resource sharing, objectives, areas and operationalities of resource sharing, IFLA guidelines for resource sharing and document delivery

Overview of library and Information Networks: Need and objectives of library networks, OCLC, INFLIBNET, DELNET, ADINET

Network-based Information Services

Management of Library and Information Networks

Unit-3

Internet Technology

What is Internet?

Internet Services

Intranet, Extranet and Internet

Unit-4

Web 2.0: Web 2.0 technologies and services; RSS, podcasts and vodcasts, blogs, wikis, social networks.

Web 3.0: Components and services: Semantic Web, Semantic Wiki, 3.0 Web, Artificial Intelligence.

Essential Readings

Andrew S. Tanenbaum(2002). *Computer Networks.* New Delhi: Prentice hall.

Black, U. (1992). *TCP/IP and Related Protocols.* New York: Mc.Graw Hill.

Dawson, A. (1997). *The Internet for Library and Information Professionals.* London: Library Association

Gilbert Held (1998). *Data Communications Networking Devices.* New Delhi: Wiley

Mehta, Subhash. (1996). *Understanding and Using Internet.* New Delhi, Ess Ess Publications.

Miller, M.A. (1991) *Internetworking: A Guide to Network Communications LAN to LAN; LAN to WAN.* New York: M&T Books.

Prasad, K N. (1998) *Intellectual property rights in information systems, networks and services in India.* Chennai: Ranganathan Center for Information Studies.

Radia Perlman (1999). *Interconnections, Bridges, Routers, Switches, and Internetworking Protocols.* Addison Wesley.

Raina, Roshan. (1997) *Library resource sharing and networking: An approach among management schools of India,* Vikas Publishing House.

Randall, Neil. *Teach yourself the Internet in a Week.* New Delhi: Prentice-hall of India.

Sophon, D. L. (1993). *Data Network Design.* New York: McGraw Hill.

William Stallings (2006) *Data and Computer Communications.* New Delhi: Prentice Hall of India.

Further Readings

Ahmedabad Library Network (ADINET) www.alibnet.org

Consortium of University Research Libraries (CURL) http://curl.ac.uk/

Developing Library Network (DELNET) http://www.delnet.org

Gayas-ud-din (1993). *Information networks and systems in India.* In Resource Sharing and Information networks, 8(2) pp. 119-35.

India. Lok Sabha Secre., (1987) *Background to evolving National Information Policy.* New Delhi.

Joint Acedemy Network (JANET) http://www.ukerna.ac.uk/

Lahiri, Abhijit and Sunder Singh, B.G. *Experiences of Library Network Development in India.* In CALIBER-98 on the theme Information Management in Academic and Research Libraries, Bhubaneswar 4th March, 1998.

Macleod, R and Kerr, L. *Internet : Past Present and Future.* In The Electronic Library, 15 (4), 1997

Mysore Library Network (MYLIBNET) http://www.mylibnet.org

Online Computer Library Center http:www.oclc.org/

Prasad, A. R. D. *Design and Developing Academic Library Websites, Theme paper, CALIBER-2000.* Chennai, Feb 14-16, 2000.

Research Library Network http://www.rlg.org/toc/html

Paper Code: LAN-05
Paper Title: Internet Resources and Content Development

Objectives: To make students conversant with various types of Information Resources; Content Analysis and Content Development.

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Unit-1 Types of Resources (by Form)
Meta Resources: Definition, Basic Characteristics, Types, Methodology, Organisation.
Digital Library: Definition, Characteristics, Components, Technical Infrastructure, Steps involved in building a digital library.
Internet Information Resources and their evaluation.

Unit-2 Types of Resources (by Subject)
Internet Resources: Science and Technology
Internet Resources: Social Science and Humanities
Evaluation of Internet Resources
General Portals/Gateways (BUBL information service, WWW Virtual Library, Pinakes.)
Subject Portals/Gateways (BIOME, Pubmed, Infomine, lintute: Social Sciences)

Unit-3 Content Analysis
Content Analysis: Concept, Basic elements, Need, Types (Qualitative and Quantitative Analysis),and criteria for analyzing web sites & web pages.
Content Organization: concept, need & purpose, classification, cataloguing, Dublin core, Marc 21, Web indexing, subject headings, Z39.50.

Unit-4 Content Development
Context Setting, Norms and Guidelines, HTML, XML, Web based content development.
Intellectual Property Rights: Legal and ethical issues.

Essential Readings

Broadbent, Brooke. (2002) ABCs of E-Learning: Reaping the Benefits and Avoiding the Pitfalls. Jossey-Bass.

Cooke, A.A (1999) Guide to finding information on the Internet: selection, evaluation and strategies. London:Library Association Publishing.

Kaplan, Jim. The Auditor's Guide to Internet Resources (Paperback)Institute of Internal Auditors IS.

Gordon, Rachel Singer. (2001)Teaching the Internet in Libraries.

Karol, E.D. and Ferguson, P. (1995) The Whole Internet for Windows 95: User's Guide and Catalog. O'Reilly & Associate, Inc.

Kinder, R (Editor). (1994) Librarians on the Internet: Impact on reference services. New York, Hawthorn Press.

Rosenberg, Mark J. E-Learning: Strategies for Delivering Knowledge in the Digital Age. McGraw-Hill.

Sherman, Chris and Price, Gary. The Invisible Web: Uncovering Information Sources search Independent Publishers Group.

Tseng, G., Poulter, A and Hiom, D. (1997) The Library and Information Professional's Guide to the Internet. 2d ed., London: Library Association Publishing.

Wood, M, Sandra (Editor). Health Care Resources on the Internet: A Guide for Librarians and Health Care Consumers-Novemeber 1999) Haworth Press.

Further Readings

Archives & Museum Informatics. http://www.ncl-bnc.ca/ifla/II/copyright.htm

Demas, S.Et al.' The Internet and collection Development: mainstreaming selection of Internet resources. Library Resources and Technical Services, 39 (3), 1995, 275-90.

Frater, Harald and Paulissen, Dirk (1994). Multimedia Mania. Abacus: Grand Rapids, 199

Intellectual Property: Law & amp; Procedure in India. http://www.singhania.com/ip/contents.html

Perry, Paul (1994). Multimedia Developers Guide. Same Publishing: Indianapolis, 1994

Rosch, Winn L (1995).Multimedia Bible.Sams Publishing, Indianapolis,1995.
Plagiarism.org. http/www.plagiarism.org/problem.html

Wodaski, Ron (1994) Multimedia Madness.Sams Publishing:Indianapolis.1994.

Smith,C.P.(ED.)(1992) Motivation and Personality: Handbook of Thematic Content Analysis.New York: Cambridge University Press.

Paper Code: LAN-06
Paper Title: Computer Practical

Objectives: To develop computer skills and professional competencies to solve problems of libraries and information centers through practical knowledge of computers.

Instructions for the Paper-Setters / Examiners
The Paper carries 80 marks. Duration of the paper is 3 hours. There will be 2 parts, out of which Part- I is compulsory based on **viva-voce** of 20 marks. Part - II contains 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

UNIT-1

1. MS Windows

- Working with in MS WINDOWS
- Using various services /options
- File management

2. MS-OFFICE

- MS WORD
- MS EXCEL
- MS POWERPOINT
- MS ACCESS : Creating book Database & Patron Database

3. PC security, Tools & Services

- Antivirus
- Internal security
- Firewall
- Free tools ,services & their usage

UNIT-2

1. Library Automation

- Installation & Working with KOHA

2. WEB OPAC's

- Use of Regional ,National & International OPACs
- Generating Bibliographies

3. FEDRATED SEARCH

- Hands –on/ practical on open access federated search tools

UNIT-3

1. Web 2.0 & Library 2.0

- RSS feeds: Social Tagging & Bookmarking
- Blogs: Creation steps & implementing value added features
- Wikis: Creation & usage

2. Online Reference Managers

- Using/Installation of online reference managers like 2collab, Zotero

3. Online Bibliographies Makers

- Creation of bibliographies of books, Journals, websites, newspapers .
- According to common citation styles using free online resources.

UNIT-4

1. INTERNET

- Basics of Internet
- Framing search strategies in search engine/meta search engines like Google/ dogpile.
- Use of Academic Search Engine

2. NETWORKING AND RESOURCE SHARING

- File, folder, printer sharing
- Networking tools and services

3. Digital Library

- Installation & creating of Digital Library using Greenstone
- Basic customization of user Interface

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